INCH-POUND

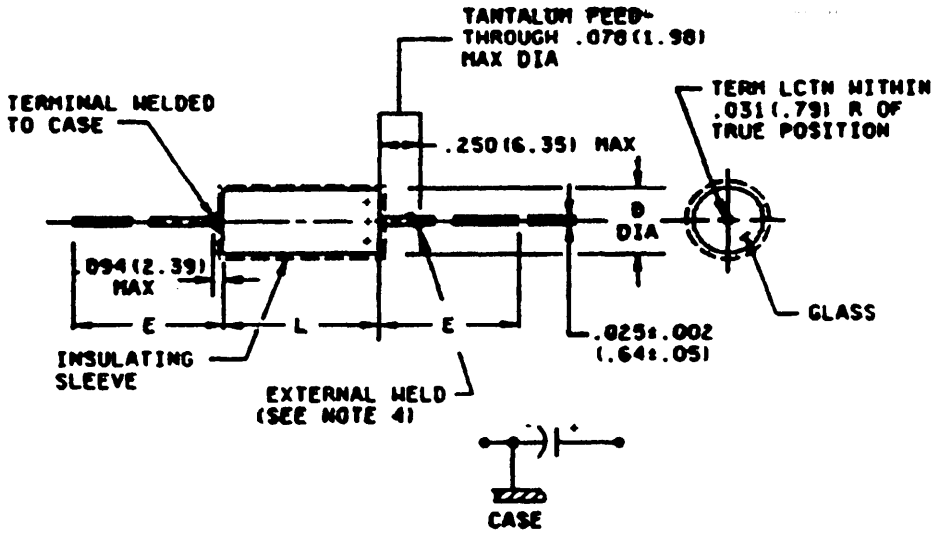
MIL-C-39006/30
10 March 1994

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR90

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39006.



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR90 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Dimensions				
Case size	Basic case		Insulated case	
	L 1/ +.031 (0.79) -.016 (0.41)	D ±.016 (0.41)	D max	E ±.250 (6.35)
T1	.453(11.51)	.188 (4.78)	.219 (5.56)	1.500(38.10)
T2	.641(16.28)	.281 (7.14)	.312 (7.92)	2.250(57.15)
T3	.766(19.46)	.375 (9.52)	.406(10.31)	2.250(57.15)
T4	1.062(26.97)	.375 (9.52)	.406(10.31)	2.250(57.15)

1/ Length of basic case sleeving shall be specified in MIL-C-39006.

FIGURE 1. Style CLR90 capacitors - Continued.

QUALIFICATION:

A manufacturer shall be considered qualified to this specification sheet by similarity if the manufacturer is qualified to style CLR79 capacitors in MIL-C-39006/22 and if the design and construction of the capacitors in this specification sheet is the same as the design and construction of the capacitors covered by MIL-C-39006/22.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tantalum, tubular, insulated.

Seal type: Hermetic (glass-to-tantalum anode). For compression type, see table 1.

Terminals: Axial-wire lead: Cathode and anode (N32, N51, or N52 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

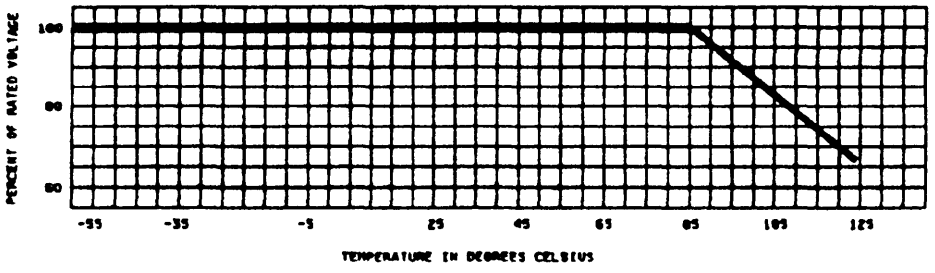


FIGURE 2. Voltage derating with temperature.

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TABLE I. Capacitor characteristics.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max) +25°C	DC leakage (max) +85°C, +125°C	Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ nonconcession seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ concession seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
6	30	±20	1	2	4.5	1.99	4	6.9	160	-40	+10.5	+12	820	T1	0001-	0221-	0441-	0661-	1001-	2011-	3061-	4061-
"	30	±10	"	"	"	"	"	"	160	"	"	"	820	"	0002-	0222-	0442-	0662-	1002-	2012-	3062-	4062-
"	30	±5	"	"	"	"	"	"	160	"	"	"	820	"	0003-	0223-	0443-	0663-	1003-	2013-	3063-	4063-
"	68	±20	"	"	7.5	1.58	"	"	80	"	+14	+16	960	"	0004-	0224-	0444-	0664-	1004-	2014-	3064-	4064-
"	68	±10	"	"	"	"	"	"	80	"	"	"	960	"	0005-	0225-	0445-	0665-	1005-	2015-	3065-	4065-
"	68	±5	"	"	"	"	"	"	80	"	"	"	960	"	0006-	0226-	0446-	0666-	1006-	2016-	3066-	4066-
"	140	±20	"	3	10.5	0.99	"	"	46	"	"	"	1280	T2	0007-	0227-	0447-	0667-	1007-	2017-	3067-	4067-
"	140	±10	"	"	"	"	"	"	46	"	"	"	1280	"	0008-	0228-	0448-	0668-	1008-	2018-	3068-	4068-
"	140	±5	"	"	"	"	"	"	46	"	"	"	1280	"	0009-	0229-	0449-	0669-	1009-	2019-	3069-	4069-
"	270	±20	"	6.5	22.5	1.11	"	"	25	-44	+17.5	+20	1373	"	0010-	0230-	0450-	0670-	1010-	2020-	3070-	4070-
"	270	±10	"	"	"	"	"	"	25	"	"	"	1375	"	0011-	0231-	0451-	0671-	1011-	2021-	3071-	4071-
"	270	±5	"	"	"	"	"	"	25	"	"	"	1375	"	0012-	0232-	0452-	0672-	1012-	2022-	3072-	4072-
"	330	±20	2	7.9	18	0.73	"	"	20	"	+14	+16	1680	T3	0013-	0233-	0453-	0673-	1013-	2023-	3073-	4073-
"	330	±10	"	"	"	"	"	"	20	"	"	"	1680	"	0014-	0234-	0454-	0674-	1014-	2024-	3074-	4074-
"	330	±5	"	"	"	"	"	"	20	"	"	"	1680	"	0015-	0235-	0455-	0675-	1015-	2025-	3075-	4075-
"	560	±20	"	13	27.5	0.65	"	"	15	-84	+17.5	+20	1900	"	0016-	0236-	0456-	0676-	1016-	2026-	3076-	4076-
"	560	±10	"	"	"	"	"	"	15	"	"	"	1900	"	0017-	0237-	0457-	0677-	1017-	2027-	3077-	4077-
"	560	±5	"	"	"	"	"	"	15	"	"	"	1900	"	0018-	0238-	0458-	0678-	1018-	2028-	3078-	4078-
"	1200	±20	3	14	45	0.50	"	"	28	-88	+25	+25	2265	T4	0019-	0239-	0459-	0679-	1019-	2029-	3079-	4079-
"	1200	±10	3	14	"	"	"	"	28	-88	+25	+25	2265	T4	0020-	0240-	0460-	0680-	1020-	2030-	3080-	4080-
8	25	±20	1	2	3.75	1.99	5	9.2	168	-40	+10.5	+12	820	T1	0021-	0241-	0461-	0681-	1021-	2031-	3081-	4081-
"	25	±10	"	"	"	"	"	"	168	"	"	+12	820	"	0022-	0242-	0462-	0632-	1022-	2032-	3082-	4082-
"	25	±5	"	"	"	"	"	"	168	"	"	+12	820	"	0023-	0243-	0463-	0683-	1023-	2033-	3083-	4083-
"	56	±20	"	"	7	1.66	"	"	59	"	+14	+16	900	"	0024-	0244-	0464-	0684-	1024-	2034-	3084-	4084-
"	56	±10	"	"	"	"	"	"	59	"	"	+16	900	"	0025-	0245-	0465-	0685-	1025-	2035-	3085-	4085-
"	56	±5	"	"	"	"	"	"	59	"	"	+16	900	"	0026-	0246-	0466-	0686-	1026-	2036-	3086-	4086-
"	120	±20	"	"	10	1.11	"	"	58	-64	+17.5	+20	1220	T2	0027-	0247-	0467-	0687-	1027-	2037-	3087-	4087-
"	120	±10	"	"	"	"	"	"	58	"	"	"	1220	"	0028-	0248-	0468-	0688-	1028-	2038-	3088-	4088-
"	120	±5	"	"	"	"	"	"	58	"	"	"	1220	"	0029-	0249-	0469-	0689-	1029-	2039-	3089-	4089-
"	220	±20	"	"	18.5	1.12	"	"	38	"	"	"	1370	"	0030-	0250-	0470-	0690-	1030-	2040-	3070-	4090-
"	220	±10	"	"	"	"	"	"	38	"	"	"	1370	"	0031-	0251-	0471-	0691-	1031-	2041-	3071-	4091-
"	220	±5	"	"	"	"	"	"	38	"	"	"	1370	"	0032-	0252-	0472-	0692-	1032-	2042-	3072-	4092-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ nonconpression seal				PIN M39006/30- 4/ 5/ compression seal			
			+25°C	+85°C, +125°C						-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)				Failure rate level (%/1,000 hr.)			
															M(1.0)	P(0.1)	R(0.01)	S(0.001)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
8	290	±20	2	6	17	0.78	5	9.2	25	-64	+17.5	+20	1770	T3	0033-	0253-	0473-	0693-	1033-	2043-	3073-	4093-
"	290	±10	"	"	"	"	"	"	"	"	"	"	1770	"	0034-	0254-	0474-	0694-	1034-	2044-	3074-	4094-
"	290	±5	"	"	"	"	"	"	"	"	"	"	1770	"	0035-	0255-	0475-	0695-	1035-	2045-	3075-	4095-
"	430	±20	"	14	23	0.71	"	"	"	"	"	"	1825	"	0036-	0256-	0476-	0696-	1036-	2046-	3076-	4096-
"	430	±10	"	"	"	"	"	"	"	"	"	"	1825	"	0037-	0257-	0477-	0697-	1037-	2047-	3077-	4097-
"	430	±5	"	"	"	"	"	"	"	"	"	"	1825	"	0038-	0258-	0478-	0698-	1038-	2048-	3078-	4098-
"	850	±20	4	16	30	0.47	"	"	22	-80	+25	+25	2330	T4	0039-	0259-	0479-	0699-	1039-	2049-	3079-	4099-
"	850	±10	4	16	"	"	"	"	22	-80	+25	+25	2330	T4	0040-	0260-	0480-	0700-	1040-	2050-	3080-	4100-
10	20	±20	1	2	3	1.99	7	11.5	175	-32	+10.5	+12	620	T1	0041-	0261-	0481-	0701-	1041-	2051-	3081-	4101-
"	20	±10	"	"	"	"	"	"	175	"	"	"	620	"	0042-	0262-	0482-	0702-	1042-	2052-	3082-	4102-
"	20	±5	"	"	"	"	"	"	175	"	"	"	620	"	0043-	0263-	0483-	0703-	1043-	2053-	3083-	4103-
"	47	±20	"	"	6.5	1.84	"	"	100	-36	+14	+16	855	"	0044-	0264-	0484-	0704-	1044-	2054-	3084-	4104-
"	47	±10	"	"	"	"	"	"	100	"	"	"	855	"	0045-	0265-	0485-	0705-	1045-	2055-	3085-	4105-
"	47	±5	"	"	"	"	"	"	100	"	"	"	855	"	0046-	0266-	0486-	0706-	1046-	2056-	3086-	4106-
"	100	±20	"	4	7.5	0.99	"	"	60	"	"	"	1200	T2	0047-	0267-	0487-	0707-	1047-	2057-	3087-	4107-
"	100	±10	"	"	"	"	"	"	60	"	"	"	1200	"	0048-	0268-	0488-	0708-	1048-	2058-	3088-	4108-
"	100	±5	"	"	"	"	"	"	60	"	"	"	1200	"	0049-	0269-	0489-	0709-	1049-	2059-	3089-	4109-
"	180	±20	"	7	15	1.11	"	"	40	"	"	"	1365	"	0050-	0270-	0490-	0710-	1050-	2060-	3090-	4110-
"	180	±10	"	"	"	"	"	"	40	"	"	"	1365	"	0051-	0271-	0491-	0711-	1051-	2061-	3091-	4111-
"	180	±5	"	"	"	"	"	"	40	"	"	"	1365	"	0052-	0272-	0492-	0712-	1052-	2062-	3092-	4112-
"	250	±20	2	10	"	0.80	"	"	30	-40	"	"	1720	T3	0053-	0273-	0493-	0713-	1053-	2063-	3093-	4113-
"	250	±10	"	"	"	"	"	"	30	"	"	"	1720	"	0054-	0274-	0494-	0714-	1054-	2064-	3094-	4114-
"	250	±5	"	"	"	"	"	"	30	"	"	"	1720	"	0055-	0275-	0495-	0715-	1055-	2065-	3095-	4115-
"	390	±20	"	16	22	0.75	"	"	25	-64	+17.5	+20	1800	"	0056-	0276-	0496-	0716-	1056-	2064-	3096-	4116-
"	390	±10	"	"	"	"	"	"	25	"	"	"	1800	"	0057-	0277-	0497-	0717-	1057-	2067-	3097-	4117-
"	390	±5	"	"	"	"	"	"	25	"	"	"	1800	"	0058-	0278-	0498-	0718-	1058-	2068-	3098-	4118-
"	750	±20	4	"	25	0.44	"	"	23	-80	+25	+25	2340	T4	0059-	0279-	0499-	0719-	1059-	2069-	3099-	4119-
"	750	±10	4	"	"	"	"	"	23	-80	"	+25	2340	T4	0060-	0280-	0500-	0720-	1060-	2070-	3100-	4120-
15	15	±20	1	2	2.5	2.21	10	17.2	155	-24	+10.5	+12	780	T1	0061-	0281-	0501-	0721-	1061-	2071-	3101-	4121-
"	15	±10	"	"	"	"	"	"	155	"	"	"	780	"	0062-	0282-	0502-	0722-	1062-	2072-	3102-	4122-
"	15	±5	"	"	"	"	"	"	155	"	"	"	780	"	0063-	0283-	0503-	0723-	1063-	2073-	3103-	4123-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max) +25°C	DC leakage (max) +85°C, +125°C	Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ noncompression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
15	33	±20	1	2	5	2.01	10	17.2	90	-20	+14	+16	820	T1	0064-	0284-	0504-	0724-	1064-	2074-	3104-	4126-
"	33	±10	"	"	"	"	"	"	90	"	"	"	820	"	0065-	0285-	0505-	0725-	1065-	2075-	3105-	4125-
"	33	±5	"	"	"	"	"	"	90	"	"	"	820	"	0064-	0286-	0506-	0726-	1064-	2076-	3106-	4126-
"	70	±20	"	4	6.5	1.23	"	"	75	"	"	"	1150	T2	0067-	0287-	0507-	0727-	1067-	2077-	3107-	4127-
"	70	±10	"	"	"	"	"	"	75	"	"	"	1150	"	0068-	0288-	0508-	0728-	1068-	2078-	3108-	4128-
"	70	±5	"	"	"	"	"	"	75	"	"	"	1150	"	0069-	0289-	0509-	0729-	1069-	2079-	3109-	4129-
"	120	±20	"	7	9	0.99	"	"	50	"	+17.5	+20	1450	"	0070-	0290-	0510-	0730-	1070-	2080-	3110-	4130-
"	120	±10	"	"	"	"	"	"	50	"	"	"	1450	"	0071-	0291-	0511-	0731-	1071-	2081-	3111-	4131-
"	120	±5	"	"	"	"	"	"	50	"	"	"	1450	"	0072-	0292-	0512-	0732-	1072-	2082-	3112-	4132-
"	170	±20	2	10	12.5	0.98	"	"	35	-32	+14	+16	1480	T3	0073-	0293-	0513-	0733-	1073-	2083-	3113-	4133-
"	170	±10	"	"	"	"	"	"	35	"	"	"	1480	"	0074-	0294-	0514-	0734-	1074-	2084-	3114-	4134-
"	170	±5	"	"	"	"	"	"	35	"	"	"	1480	"	0075-	0295-	0515-	0735-	1075-	2085-	3115-	4135-
"	270	±20	"	16	16	0.79	"	"	30	-56	+17.5	+20	1740	"	0076-	0296-	0516-	0736-	1076-	2086-	3116-	4136-
"	270	±10	"	"	"	"	"	"	30	"	"	"	1740	"	0077-	0297-	0517-	0737-	1077-	2087-	3117-	4137-
"	270	±5	"	"	"	"	"	"	30	"	"	"	1740	"	0078-	0298-	0518-	0738-	1078-	2088-	3118-	4138-
"	540	±20	6	24	20	0.49	10	17.2	25	-88	+25	+25	2330	T4	0079-	0299-	0519-	0739-	1079-	2089-	3119-	4139-
"	540	±10	6	24	"	"	10	17.2	25	-88	+25	+26	2330	T4	0080-	0300-	0520-	0740-	1080-	2090-	3120-	4140-
25	10	±20	1	2	2	2.66	15	28.8	220	-14	+8	+9	715	T1	0081-	0301-	0521-	0741-	1081-	2091-	3121-	4141-
"	10	±10	"	"	"	"	"	"	220	"	+8	"	715	"	0082-	0302-	0522-	0742-	1082-	2092-	3122-	4142-
"	10	±5	"	"	"	"	"	"	220	"	+8	"	715	"	0083-	0303-	0523-	0743-	1083-	2093-	3123-	4143-
"	22	±20	"	"	3.3	1.99	"	"	140	-28	+10.5	+12	825	"	0084-	0304-	0524-	0744-	1084-	2094-	3124-	4144-
"	22	±10	"	"	"	"	"	"	140	"	"	"	825	"	0085-	0305-	0525-	0745-	1085-	2095-	3125-	4145-
"	22	±5	"	"	"	"	"	"	140	"	"	"	825	"	0086-	0306-	0526-	0746-	1086-	2096-	3126-	4146-
"	50	±20	"	"	5.5	1.46	"	"	70	-28	+13	+15	1130	T2	0087-	0307-	0527-	0747-	1087-	2097-	3127-	4147-
"	50	±10	"	"	"	"	"	"	70	"	"	"	1130	"	0088-	0308-	0528-	0748-	1088-	2098-	3128-	4148-
"	50	±5	"	"	"	"	"	"	70	"	"	"	1130	"	0089-	0309-	0529-	0749-	1089-	2099-	3129-	4149-
"	100	±20	"	10	7.5	0.99	"	"	50	"	"	"	1435	"	0090-	0310-	0530-	0750-	1090-	2100-	3130-	4150-
"	100	±10	"	"	"	"	"	"	50	"	"	"	1435	"	0091-	0311-	0531-	0751-	1091-	2101-	3131-	4151-
"	100	±5	"	"	"	"	"	"	50	"	"	"	1435	"	0092-	0312-	0532-	0752-	1092-	2102-	3132-	4152-
"	120	±20	2	6	10.5	1.16	"	"	38	-32	"	"	1450	T3	0093-	0313-	0533-	0753-	1093-	2103-	3133-	4153-
"	120	±10	"	"	"	"	"	"	38	"	"	"	1450	"	0094-	0314-	0534-	0754-	1094-	2104-	3134-	4154-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ nonexpression seal				PIN M39006/30- 4/ 5/ expression seal			
			+25°C	+85°C, +125°C						-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)				Failure rate level (%/1,000 hr.)			
															M(1.0)	P(0.1)	R(0.01)	S(0.001)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
25	120	±5	2	6	10.5	1.16	15	28.8	38	-32	+13	+15	1450	T3	0095-	0315-	0535-	0755-	1015-	2105-	3135-	4155-
"	180	±20	"	18	13	0.96	"	"	32	-48	"	"	1525	"	0096-	0316-	0536-	0756-	1016-	2106-	3136-	4156-
"	180	±10	"	"	"	"	"	"	32	"	"	"	1525	"	0097-	0317-	0537-	0757-	1017-	2107-	3137-	4157-
"	180	±5	"	"	"	"	"	"	32	"	"	"	1525	"	0098-	0318-	0538-	0758-	1018-	2108-	3138-	4158-
"	350	±20	7	28	17.5	0.67	"	"	24	-70	+25	+25	1970	T4	0099-	0319-	0539-	0759-	1019-	2109-	3139-	4159-
"	350	±10	7	28	"	"	"	"	24	-70	+25	+25	1970	T4	0100-	0320-	0540-	0760-	1110-	2110-	3140-	4160-
30	8	±20	1	2	2	3.32	20	34.5	275	-16	+8	+12	640	T1	0101-	0321-	0541-	0761-	1111-	2111-	3141-	4161-
"	8	±10	"	"	"	"	"	"	275	"	+8	"	640	"	0102-	0322-	0542-	0762-	1112-	2112-	3142-	4162-
"	8	±5	"	"	"	"	"	"	275	"	+8	"	640	"	0103-	0323-	0543-	0763-	1113-	2113-	3143-	4163-
"	15	±20	"	"	2.5	2.21	"	"	175	-28	+10.5	"	780	"	0104-	0324-	0544-	0764-	1114-	2114-	3144-	4164-
"	15	±10	"	"	"	"	"	"	175	"	"	"	780	"	0105-	0325-	0545-	0765-	1115-	2115-	3145-	4165-
"	15	±5	"	"	"	"	"	"	175	"	"	"	780	"	0106-	0326-	0546-	0766-	1116-	2116-	3146-	4166-
"	40	±20	"	5	5	1.66	"	"	85	-24	"	"	1120	T2	0107-	0327-	0547-	0767-	1117-	2117-	3147-	4167-
"	40	±10	"	"	"	"	"	"	85	"	"	"	1120	"	0108-	0328-	0548-	0768-	1118-	2118-	3148-	4168-
"	40	±5	"	"	"	"	"	"	85	"	"	"	1120	"	0109-	0329-	0549-	0769-	1119-	2119-	3149-	4169-
"	68	±20	"	8	6.5	1.27	"	"	60	"	+13	+15	1285	"	0110-	0330-	0550-	0770-	1110-	2120-	3150-	4170-
"	68	±10	"	"	"	"	"	"	60	"	"	"	1285	"	0111-	0331-	0551-	0771-	1111-	2121-	3151-	4171-
"	68	±5	"	"	"	"	"	"	60	"	"	"	1285	"	0112-	0332-	0552-	0772-	1112-	2122-	3152-	4172-
"	100	±20	2	12	8.5	1.13	"	"	40	-28	+10.5	+12	1450	T3	0113-	0333-	0553-	0773-	1113-	2123-	3153-	4173-
"	100	±10	"	"	"	"	"	"	40	"	"	"	1450	"	0114-	0334-	0554-	0774-	1114-	2124-	3154-	4174-
"	100	±5	"	"	"	"	"	"	40	"	"	"	1450	"	0115-	0335-	0555-	0775-	1115-	2125-	3155-	4175-
"	150	±20	2	18	11.5	1.02	20	34.5	35	-48	+13	+15	1525	T3	0116-	0336-	0556-	0776-	1116-	2126-	3156-	4176-
"	150	±10	"	"	"	"	"	"	35	-48	+13	+15	1525	"	0117-	0337-	0557-	0777-	1117-	2127-	3157-	4177-
"	150	±5	"	"	"	"	"	"	35	-48	+13	+15	1525	"	0118-	0338-	0558-	0778-	1118-	2128-	3158-	4178-
"	300	±20	8	32	15.5	0.69	"	"	25	-60	+25	+25	1950	T4	0119-	0339-	0559-	0779-	1119-	2129-	3159-	4179-
"	300	±10	8	32	"	"	"	"	25	-60	+25	+25	1950	T4	0120-	0340-	0560-	0780-	1120-	2130-	3160-	4180-
50	5	±20	1	2	1.5	3.98	30	57.5	400	-16	+5	+6	580	T1	0121-	0341-	0561-	0781-	1121-	2131-	3161-	4181-
"	5	±10	"	"	"	"	"	"	400	-16	+5	+6	580	"	0122-	0342-	0562-	0782-	1122-	2132-	3162-	4182-
"	5	±5	"	"	"	"	"	"	400	-16	+5	+6	580	"	0123-	0343-	0563-	0783-	1123-	2133-	3163-	4183-
"	10	±20	"	"	2	2.66	"	"	250	-24	+8	+9	715	"	0124-	0344-	0564-	0784-	1124-	2134-	3164-	4184-
"	10	±10	"	"	"	"	"	"	250	-24	+8	+9	715	"	0125-	0345-	0565-	0785-	1125-	2135-	3165-	4185-
"	10	±5	"	"	"	"	"	"	250	-24	+8	+9	715	"	0126-	0346-	0566-	0786-	1126-	2136-	3166-	4186-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max) +25°C	DC leakage (max) +85°C, +125°C	Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ noncompression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
50	25	±20	1	5	4	2.13	30	57.5	95	-20	+10.5	+12	1005	T2	0127-	0347-	0567-	0787-	1127-	2137-	3167-	4187-
"	25	±10	"	"	"	"	"	"	95	-20	+10.5	+12	1005	"	0128-	0348-	0568-	0788-	1128-	2138-	3168-	4188-
"	25	±5	"	"	"	"	"	"	95	-20	+10.5	+12	1005	"	0129-	0349-	0569-	0789-	1129-	2139-	3169-	4189-
"	47	±20	"	9	5.5	1.56	"	"	70	-20	+13	+15	1155	"	0130-	0350-	0570-	0790-	1130-	2140-	3170-	4190-
"	47	±10	"	"	"	"	30	57.5	70	-20	+13	+15	1155	T2	0131-	0351-	0571-	0791-	1131-	2141-	3171-	4191-
"	47	±5	"	"	"	"	"	"	70	-20	+13	+15	1155	T2	0132-	0352-	0572-	0792-	1132-	2142-	3172-	4192-
"	60	±20	2	12	6	1.33	"	"	45	-16	+10.5	+12	1335	T3	0133-	0353-	0573-	0793-	1133-	2143-	3173-	4193-
"	60	±10	"	"	"	"	"	"	"	-16	+10.5	+12	1335	"	0134-	0354-	0574-	0794-	1134-	2144-	3174-	4194-
"	60	±5	"	"	"	"	"	"	"	-16	+10.5	+12	1335	"	0135-	0355-	0575-	0795-	1135-	2145-	3175-	4195-
"	82	±20	"	16	7.5	1.22	"	"	"	-32	+13	+15	1408	"	0136-	0356-	0576-	0796-	1136-	2146-	3176-	4196-
"	82	±10	"	"	"	"	"	"	"	-32	+13	+15	1408	"	0137-	0357-	0577-	0797-	1137-	2147-	3177-	4197-
"	82	±5	"	"	"	"	"	"	"	-32	+13	+15	1408	"	0138-	0358-	0578-	0798-	1138-	2148-	3178-	4198-
"	160	±20	8	32	8.5	0.71	"	"	27	-30	+25	+25	1900	T4	0139-	0359-	0579-	0799-	1139-	2149-	3179-	4199-
"	160	±10	8	32	"	"	"	"	27	-30	+25	+25	1900	T4	0140-	0360-	0580-	0800-	1140-	2150-	3180-	4200-
60	4	±20	1	2	1.4	4.65	40	69.0	550	-16	+5	+6	525	T1	0141-	0361-	0581-	0801-	1141-	2151-	3181-	4201-
"	4	±10	"	"	"	"	"	"	550	-16	+5	+6	525	"	0142-	0362-	0582-	0802-	1142-	2152-	3182-	4202-
"	4	±5	"	"	"	"	"	"	550	-16	+5	+6	525	"	0143-	0363-	0583-	0803-	1143-	2153-	3183-	4203-
"	8.2	±20	"	"	2	3.24	"	"	275	-24	+8	+9	625	"	0144-	0364-	0584-	0804-	1144-	2154-	3184-	4204-
"	8.2	±10	"	"	"	"	"	"	275	-24	+8	+9	625	"	0145-	0365-	0585-	0805-	1145-	2155-	3185-	4205-
"	8.2	±5	"	"	"	"	"	"	275	-24	+8	+9	625	"	0146-	0366-	0586-	0806-	1146-	2156-	3186-	4206-
"	20	±20	"	5	3.5	2.32	"	"	105	-16	+10.5	+12	930	T2	0147-	0367-	0587-	0807-	1147-	2157-	3187-	4207-
"	20	±10	"	"	"	"	"	"	105	-16	"	"	930	"	0148-	0368-	0588-	0808-	1148-	2158-	3188-	4208-
"	20	±5	"	"	"	"	"	"	105	-16	"	"	930	"	0149-	0369-	0589-	0809-	1149-	2159-	3189-	4209-
"	39	±20	"	9	5	1.70	"	"	90	-28	"	"	1110	"	0150-	0370-	0590-	0810-	1150-	2160-	3190-	4210-
"	39	±10	"	"	"	"	"	"	90	-28	"	"	1110	"	0151-	0371-	0591-	0811-	1151-	2161-	3191-	4211-
"	39	±5	"	"	"	"	"	"	90	-28	"	"	1110	"	0152-	0372-	0592-	0812-	1152-	2162-	3192-	4212-
"	50	±20	2	12	"	1.33	"	"	50	-16	"	"	1330	T3	0153-	0373-	0593-	0813-	1153-	2163-	3193-	4213-
"	50	±10	"	"	"	"	"	"	50	-16	"	"	1330	"	0154-	0374-	0594-	0814-	1154-	2164-	3194-	4214-
"	50	±5	"	"	"	"	"	"	50	-16	"	"	1330	"	0155-	0375-	0595-	0815-	1155-	2165-	3195-	4215-
"	68	±20	2	16	6.5	1.27	40	69.0	50	-32	+10.5	+12	1365	T3	0156-	0376-	0596-	0816-	1156-	2166-	3196-	4216-
"	68	±10	"	"	"	"	"	"	50	-32	+10.5	+12	1365	"	0157-	0377-	0597-	0817-	1157-	2167-	3197-	4217-
"	68	±5	"	"	"	"	"	"	50	-32	+10.5	+12	1365	"	0158-	0378-	0598-	0818-	1158-	2168-	3198-	4218-

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max) +25°C	DC leakage (max) +85°C, +125°C	Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ noncompression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
60	140	±20	8	32	8	0.76	40	69.0	28	-40	+20	+20	1858	T4	0159-	0379-	0599-	0819-	1159-	2160-	3199-	4219-
60	140	±10	8	32	8	"	40	69.0	28	-40	+20	+20	1858	T4	0160-	0380-	0600-	0820-	1160-	2170-	3200-	4220-
75	3.5	±20	1	2	1.25	4.74	50	86.2	650	-16	+5	+4	525	T1	0161-	0381-	0601-	0821-	1161-	2171-	3201-	4221-
"	3.5	±10	"	"	"	"	"	"	650	-16	+5	+4	525	"	0162-	0382-	0602-	0822-	1162-	2172-	3202-	4222-
"	3.5	±5	"	"	"	"	"	"	650	-16	+5	+4	525	"	0163-	0383-	0603-	0823-	1163-	2173-	3203-	4223-
"	6.8	±20	"	"	1.75	3.42	"	"	300	-20	+6	+9	610	"	0164-	0384-	0604-	0824-	1164-	2174-	3204-	4224-
"	6.8	±10	"	"	"	"	"	"	300	-20	"	"	610	"	0165-	0385-	0605-	0825-	1165-	2175-	3205-	4225-
"	6.8	±5	"	"	"	"	"	"	300	-20	"	"	610	"	0166-	0386-	0606-	0826-	1166-	2176-	3206-	4226-
"	15	±20	"	5	3	2.66	"	"	150	-14	"	"	890	T2	0167-	0387-	0607-	0827-	1167-	2177-	3207-	4227-
"	15	±10	"	"	"	"	"	"	150	-14	"	"	890	"	0168-	0388-	0608-	0828-	1168-	2178-	3208-	4228-
"	15	±5	"	"	"	"	"	"	150	-14	"	"	890	"	0169-	0389-	0609-	0829-	1169-	2179-	3209-	4229-
"	33	±20	"	10	5	2.01	"	"	90	-24	+10.5	+15	1008	"	0170-	0390-	0610-	0830-	1170-	2180-	3210-	4230-
"	33	±10	"	"	"	"	"	"	90	-24	"	"	1008	"	0171-	0391-	0611-	0831-	1171-	2181-	3211-	4231-
"	33	±5	"	"	"	"	"	"	90	-24	"	"	1008	"	0172-	0392-	0612-	0832-	1172-	2182-	3212-	4232-
"	40	±20	2	12	4.5	1.50	"	"	80	-16	"	+12	1258	T3	0173-	0393-	0613-	0833-	1173-	2183-	3213-	4233-
"	40	±10	"	"	"	"	"	"	"	-16	"	"	1258	"	0174-	0394-	0614-	0834-	1174-	2184-	3214-	4234-
"	40	±5	"	"	"	"	"	"	"	-16	"	"	1258	"	0175-	0395-	0615-	0835-	1175-	2185-	3215-	4235-
"	56	±20	"	17	5.5	1.31	"	"	"	-28	"	+15	1335	"	0176-	0396-	0616-	0836-	1176-	2186-	3216-	4236-
"	56	±10	"	"	"	"	"	"	"	-28	"	"	1335	"	0177-	0397-	0617-	0837-	1177-	2187-	3217-	4237-
"	56	±5	"	"	"	"	"	"	"	-28	"	"	1335	"	0178-	0398-	0618-	0838-	1178-	2188-	3218-	4238-
"	110	±20	9	36	6	0.73	"	"	29	-35	+20	+20	1850	T4	0179-	0399-	0619-	0839-	1179-	2189-	3219-	4239-
"	110	±10	9	36	"	"	"	"	29	-35	+20	+20	1850	T4	0180-	0400-	0620-	0840-	1180-	2190-	3220-	4240-
100	2.5	±20	1	2	1	5.31	65	115	950	-16	+7	+8	505	T1	0181-	0401-	0621-	0841-	1181-	2191-	3221-	4241-
"	2.5	±10	"	"	"	"	"	"	950	"	"	"	505	"	0182-	0402-	0622-	0842-	1182-	2192-	3222-	4242-
"	2.5	±5	"	"	"	"	"	"	950	"	"	"	505	"	0183-	0403-	0623-	0843-	1183-	2193-	3223-	4243-
"	4.7	±20	"	"	1.5	4.24	"	"	500	"	"	"	565	"	0184-	0404-	0624-	0844-	1184-	2194-	3224-	4244-
"	4.7	±10	"	"	"	"	"	"	500	"	"	"	565	"	0185-	0405-	0625-	0845-	1185-	2195-	3225-	4245-
"	4.7	±5	"	"	"	"	"	"	500	"	"	"	565	"	0186-	0406-	0626-	0846-	1186-	2196-	3226-	4246-
"	11	±20	"	4	2.5	3.02	"	"	200	"	+8	"	835	T2	0187-	0407-	0627-	0847-	1187-	2197-	3227-	4247-
"	11	±10	"	"	"	"	"	"	200	"	"	"	835	"	0188-	0408-	0628-	0848-	1188-	2198-	3228-	4248-
"	11	±5	"	"	"	"	"	"	200	"	"	"	835	"	0189-	0409-	0629-	0849-	1189-	2199-	3229-	4249-

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max) +25°C	DC leakage (max) +85°C, +125°C	Dissipation factor (max)	ESR at +25°C (max) 2/	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	Max 85°C 40 kHz ripple current	Case size	PIN M39006/30- 4/ 5/ noncompression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal: Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	Ohms	V dc	V dc	Ohms	%	%	%	mA rms									
100	22	±20	1	9	3.75	2.26	65	115	100	-16	+8	+8	965	T2	0190-	0410-	0630-	0850-	1190-	2200-	3230-	4250-
"	22	±10	"	"	"	"	"	"	100	"	"	"	965	"	0191-	0411-	0631-	0851-	1191-	2201-	3231-	4251-
"	22	±5	"	"	"	"	"	"	100	"	"	"	965	"	0192-	0412-	0632-	0852-	1192-	2202-	3232-	4252-
"	30	±20	2	12	3.5	1.55	"	"	80	"	"	"	1240	T3	0193-	0413-	0633-	0853-	1193-	2203-	3233-	4253-
"	30	±10	"	"	"	"	"	"	80	"	"	"	1240	"	0194-	0414-	0634-	0854-	1194-	2204-	3234-	4254-
"	30	±5	"	"	"	"	"	"	80	"	"	"	1240	"	0195-	0415-	0635-	0855-	1195-	2205-	3235-	4255-
"	43	±20	"	17	4.25	1.31	"	"	70	-20	"	"	1335	"	0196-	0416-	0636-	0856-	1196-	2206-	3236-	4256-
"	43	±10	"	"	"	"	"	"	70	-20	+8	+8	1335	"	0197-	0417-	0637-	0857-	1197-	2207-	3237-	4257-
"	43	±5	"	"	"	"	"	"	70	-20	+8	+8	1335	"	0198-	0418-	0638-	0858-	1198-	2208-	3238-	4258-
"	86	±20	9	36	5	0.77	"	"	30	-25	+15	+15	1800	T4	0199-	0419-	0639-	0859-	1199-	2209-	3239-	4259-
"	86	±10	9	36	"	"	"	"	30	-25	+15	+15	1800	T4	0200-	0420-	0640-	0860-	1200-	2210-	3240-	4260-
125	1.7	±20	1	2	1	7.81	85	144	1250	-16	+7	+8	415	T1	0201-	0421-	0641-	0861-	1201-	2211-	3241-	4261-
"	1.7	±10	"	"	"	"	"	"	1250	"	"	"	415	"	0202-	0422-	0642-	0862-	1202-	2212-	3242-	4262-
"	1.7	±5	"	"	"	"	"	"	1250	"	"	"	415	"	0203-	0423-	0643-	0863-	1203-	2213-	3243-	4263-
"	3.6	±20	"	"	1.35	4.98	"	"	600	"	"	"	520	"	0204-	0424-	0644-	0864-	1204-	2214-	3244-	4264-
"	3.6	±10	"	"	"	"	"	"	600	"	"	"	520	"	0205-	0425-	0645-	0865-	1205-	2215-	3245-	4265-
"	3.6	±5	"	"	"	"	"	"	600	"	"	"	520	"	0206-	0426-	0646-	0866-	1206-	2216-	3246-	4266-
"	9	±20	"	5	2.5	3.69	"	"	260	"	"	"	755	T2	0207-	0427-	0647-	0867-	1207-	2217-	3247-	4267-
"	9	±10	"	"	"	"	"	"	260	"	"	"	755	"	0208-	0428-	0648-	0868-	1208-	2218-	3248-	4268-
"	9	±5	"	"	"	"	"	"	260	"	"	"	755	"	0209-	0429-	0649-	0869-	1209-	2219-	3249-	4269-
"	14	±20	"	7	3	2.85	"	"	167	"	"	"	860	"	0210-	0430-	0650-	0870-	1210-	2220-	3250-	4270-
"	14	±10	"	"	"	"	"	"	167	"	"	"	860	"	0211-	0431-	0651-	0871-	1211-	2221-	3251-	4271-
"	14	±5	"	"	"	"	"	"	167	"	"	"	860	"	0212-	0432-	0652-	0872-	1212-	2222-	3252-	4272-
"	18	±20	2	9	2.5	1.85	"	"	129	"	"	"	1130	T3	0213-	0433-	0653-	0873-	1213-	2223-	3253-	4273-
"	18	±10	"	"	"	"	"	"	129	"	"	"	1130	"	0214-	0434-	0654-	0874-	1214-	2224-	3254-	4274-
"	18	±5	"	"	"	"	"	"	129	"	"	"	1130	"	0215-	0435-	0655-	0875-	1215-	2225-	3255-	4275-
"	25	±20	"	13	3	1.59	"	"	93	"	"	"	1200	"	0216-	0436-	0656-	0876-	1216-	2226-	3256-	4276-
"	25	±10	"	"	"	"	"	"	93	"	"	"	1200	"	0217-	0437-	0657-	0877-	1217-	2227-	3257-	4277-
"	25	±5	"	"	"	"	"	"	93	"	"	"	1200	"	0218-	0438-	0658-	0878-	1218-	2228-	3258-	4278-
"	56	±20	10	40	3.25	0.77	"	"	32	-25	+15	+15	1800	T4	0219-	0439-	0659-	0879-	1219-	2229-	3259-	4279-
"	56	±10	10	40	"	"	"	"	32	-25	+15	+15	1800	T4	0220-	0440-	0660-	0880-	1220-	2230-	3260-	4280-

See footnotes on next page.

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TABLE I. Capacitor characteristics - Continued.

1/ Reverse voltage rating at 85°C is 3 V dc, and at 125°C is 2 V dc.

2/ Maximum ESR is calculated by the following equation:

$$ESR\ (max) = \frac{Df}{2\pi fC}, \text{ where:}$$

DF = Maximum dissipation factor from table I;
f = 120 Hz;
C = Nominal capacitance.

3/ For ripple current limits at various temperatures, voltages, and frequencies, see table II.

4/ Dash number shall include the letter "H" to indicate the optional vibration and shock requirements (i.e., 51 g's random vibration, 80 g's sinusoidal vibration, and 500 g's shock) or the "-" will be deleted.

5/ Compression seal is generally referred to as single-seal construction; noncompression seal is generally referred to as double-seal construction.

Rated voltage: See table I.

Capacitance (cap) nominal (nom.): See table I.

Cap. tolerance: See table I.

Failure rate level: M, P, R, and S (see table I).

DC leakage (DCL) (maximum) (at +25°C): See table I.

Cap. (nom.): Method 305 of MIL-STD-202 (see table I).

Dissipation factor (DF) (maximum): See table I.

Seal: Method 112 of MIL-STD-202, conditions A or D, and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 g's) or condition D (500 g's), as applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's) or condition H (80 g's), as applicable.

Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-K (51 g's).

Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C).

Number of cycles: 300 for qualification and group C; 30 for group B.

DCL: 200 percent (maximum) of 25°C value of table I for qualification and group C.

DCL: 125 percent (maximum) of +25°C value of table I for group B.

ΔCap: Within ±5 percent of initial measurement.

DF: 115 percent (maximum) of initial requirement.

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TABLE II. Ripple current multipliers versus frequency, temperature, and applied peak voltage. 1/ 2/ 3/ 4/ 5/

Frequency of applied ripple current	120 Hz				800 Hz				1 kHz				10 kHz				40 kHz				100 kHz			
	Temp °C				Temp °C				Temp °C				Temp °C				Temp °C				Temp °C			
Ambient still air	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°
100%	.60	.39	---	---	.71	.43	---	---	.72	.45	---	---	.88	.55	---	---	1.0	.63	---	---	1.1	.69	---	---
90%	.60	.46	---	---	.71	.55	---	---	.72	.55	---	---	.88	.67	---	---	1.0	.77	---	---	1.1	.85	---	---
6/ 80%	.60	.52	.35	---	.71	.62	.42	---	.72	.62	.42	---	.88	.76	.52	---	1.0	.87	.59	---	1.1	.96	.65	---
70%	.60	.58	.44	---	.71	.69	.52	---	.72	.70	.52	---	.88	.85	.64	---	1.0	.97	.73	---	1.1	1.07	.80	---
≤66 2/3%	.60	.60	.46	.27	.71	.71	.55	.32	.72	.72	.55	.32	.88	.88	.68	.40	1.0	1.0	.77	.45	1.1	1.1	.85	.50

1/ At 125°C the rated voltage of the capacitors decreases to 66 2/3 percent of the 85°C rated voltage.
2/ The peak of the applied ac ripple voltage plus the applied dc voltage must not exceed the dc voltage rating of the capacitor either forward or reverse.
3/ The ripple current listed in table I represents a rating calculated by using a maximum internal temperature rise (ΔT) of 50°C at 40 kHz at 85°C ambient temperature, with a maximum peak rated voltage of 66 2/3 percent of the 85°C peak voltage rating.
4/ The maximum allowable internal temperature rise (ΔT) decreases linearly to a calculated 10°C rise at 125°C ambient.
5/ The internal temperature rise is directly proportional to the ESR of the capacitor, and ESR increases with decreasing frequency.
6/ Percentage of 85°C rated peak voltage.

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Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:

Pull: Method 211 of MIL-STD-202, condition A.

Wire-lead-bend: In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (maximum) of +25°C value of table I.

ΔCap: Within ±8 percent of initial measurement.

DF: 115 percent (maximum) of initial requirement.

Dielectric withstanding voltage (DWV):

Method 301 of MIL-STD-202. 2,000 V dc, minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Impedance: See table I.

ΔCap: See table I.

Step 3 (25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap: See table I.

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DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Reverse voltage: As specified in MIL-C-39006.

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Life at 85°C: Method 108 of MIL-STD-202.

2,000-hour (qualification):

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

10,000-hour (group B) and extended life:

DCL (85°C): 125 percent (maximum) of table I value.

DCL (25°C): See table I.

ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (maximum) of table I value.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

AC ripple life at 85°C (qualification and group C):

Test ripple current: See table I.

Test voltage: 66.67 percent of rated voltage (see table I).

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

Life at 125°C: Method 108 of MIL-STD-202.

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

2,000-hr (group C):

DCL (125°C): 125 percent (maximum) of table I value.

DCL (25°C): See table I.

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ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (maximum) of table I value.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet) (45,720.1 m).

Test voltage: Rated voltage.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition G.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Marking: In accordance with MIL-C-39006, with the following exception: The manufacturer may mark the T1 case size "H" designated capacitors as follows:

Example: M39006/
22H0001
JAN TH
12345
9415A

Voltage groups (85°C rated voltage):

I.....6 V through 15 V

II.....25 V through 75 V

III.....100 V and 125 V

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, MC, OS
Air Force - 17, 19, 99

Preparing activity:
DLA - ES

(Project 5910-1881)

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